

Each question is worth 5 points for a total of 160 points. The extra credit can get you an additional 5 points making the highest possible score 165 out of 160. Write answers CLEARLY. If I cannot determine your answer, the problem will receive a score of 0.

Problem 1

Graph the function $f(x) = 2x^2 - x - 2$ by determining whether its graph opens up or down and by finding its vertex, axis of symmetry, y-intercept, and x-intercept(s), if any.

Problem 2

Determine whether the function $f(x) = \frac{1-x^2}{2}$ is polynomial. If it is, state the degree. If it is not, explain why.

Problem 3

Let $f(x) = 4(x^2 + 1)(x - 2)^3$.

- a) List each real zero and its multiplicity.
- b) Determine whether the graph crosses or touches the x -axis at each x -intercept.
- c) Find the power function that the graph of f resembles for large values of $|x|$.

Problem 4

Find the domain of the rational function $R(x) = \frac{4x}{x-3}$.

Problem 5

Find the vertical, horizontal, and oblique asymptotes, if any, of the rational function $R(x) = \frac{x^3}{x^4-1}$.

Problem 6

Let $R(x) = \frac{3x+3}{2x+4}$.

- (1) Find the domain of the rational function.
- (2) Locate the intercepts, if any, of the graph.
- (3) Test for symmetry with respect to the y -axis, and the origin.
- (4) Locate the horizontal or oblique asymptotes.

Problem 7

Solve the inequality $x^2 - 9 < 0$.

Problem 8

List the potential rational zeros of the polynomial function $f(x) = x^5 - 6x^2 + 9x - 3$. Do not attempt to find the zeros.

Problem 9

Let $R = \{(2, 6), (-3, 6), (4, 9), (1, 10)\}$. Find (a) the inverse of R , and (b) determine if the inverse is a function.

Problem 10

Let $f(x) = \frac{2x}{3x-1}$. Find the inverse of f , the domain of f , and the range of f . Hint: use f^{-1} to find the range of f .

Problem 11

Solve the equation $3^{x^3} = 9^x$.

Problem 12

Change the exponential expression $a^2 = 1.6$ to an equivalent expression involving a logarithm.

Problem 13

Solve the equation $\log_3 x = 2$.

Problem 14

Find the exact value of the expression $\log_8 2 + \log_8 4$ without using a calculator.

Problem 15

Solve the equation $3\log_2(x - 1) + \log_2 4 = 5$.

Problem 16

Find the amount that results from investing \$600 at 5%, compounded daily, for 3 years.

Problem 17

Plot each point in the xy -plane. Tell in which quadrant or on what coordinate axis each point lies.

- a) $A = (-3, 2)$
- b) $B = (6, 0)$
- c) $C = (-2, -2)$
- d) $D = (6, 5)$
- e) $E = (0, -3)$
- f) $F = (6, -3)$

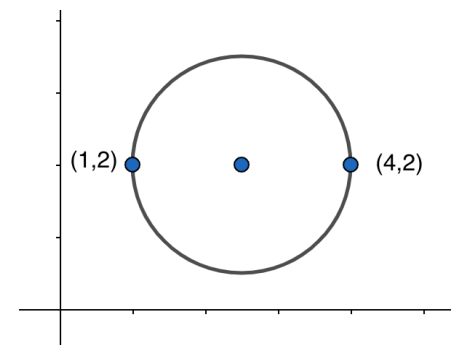
Problem 18

Plot each point in the xy -plane and form the triangle ABC . Verify that the triangle is a right triangle. Find its area.

$A = (-2, 5)$, $B = (1, 3)$, $C = (-1, 0)$

Problem 19

Find the center and the radius of the circle. Write the standard form of the equation.



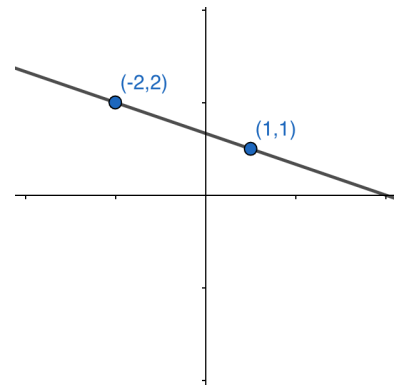
Problem 20

Find the center (h, k) and the radius r of the circle. Graph the circle.

$$x^2 + y^2 + 4x - 4y - 1 = 0$$

Problem 21

Find the equation of the line. Express your answer using either the general form or the slope-intercept form of the equation of a line.

**Problem 22**

Find the equation of the line parallel to the line $y = 2x$, and containing the point $(-1, 2)$. Express your answer using either the general form or the slope-intercept form of the equation of a line.

Problem 23

Determine whether the relation represents a function. If it's a function, state the domain and range.
 $R = \{(-2, 4), (-2, 6), (0, 3), (3, 7)\}.$

Problem 24

Find the domain of the function

$$f(x) = \frac{x}{x^2 + 1}.$$

Problem 25

Find the domain of the function

$$g(x) = \frac{x}{x^2 - 16}.$$

Problem 26

Determine algebraically whether the function $f(x) = 4x^3$ is even, odd, or neither.

Problem 27

$$\text{Let } f(x) = \begin{cases} -2x + 3 & x < 1 \\ 3x - 2 & x \geq 1 \end{cases}.$$

- a) Find the domain of f .
- b) Locate any intercepts.
- c) Graph the function.
- d) Based on the graph, find the range.

Problem 28

Find the function that is graphed after the following transformations are applied to the graph of $y = \sqrt{x}$.

- (1) Shift up two units.
- (2) Reflect about the x -axis.
- (3) Reflect about the y -axis.

Problem 29

Graph the function using the techniques of shifting, compressing, stretching, and/or reflecting. Start with the graph of the basic function (for example $y = x^2$) and show all stages. $h(x) = \sqrt{x - 2}$.

Problem 30

Let $f(x) = x - 1$ and $g(x) = 2x^2$. Find the following functions and state the domain of each.

a) $f + g$

b) $f - g$

c) $f \cdot g$

d) $\frac{f}{g}$

Problem 31

Let $P = (x, y)$ be a point on the graph of $y = x^2 - 8$.

a) Express the distance d from P to the origin as a function of x .

b) What is d if $x = 0$.

c) What is d if $x = 1$.

Problem 32

Graph the function $f(x) = x^2 + 4x + 2$ by starting with the graph $y = x^2$ and using transformations (shifting, compressing, stretching, and/or reflecting).

Extra Credit

Strontium 90 is a radioactive material that decays according to the function $A(t) = A_0 e^{-0.0244t}$, where A_0 is the initial amount present and A is the amount present after t years.

- a) What is the half-life of strontium 90?
- b) Determine how long it takes 100 grams of strontium 90 to decay to 10 grams.